

VAM Split Strategy

Technical Specification — Sourced from Client Transcripts
For Independent Audit | Ravi Mareedu Project

Tag	Meaning
■ CONFIRMED	Parameter explicitly stated by Ravi in a recorded meeting
■ FLAG	NOT SPECIFIED BY CLIENT — IFA used industry standard. Must confirm with Ravi.
■ CONFLICT	Ravi gave two different values across transcripts. Needs resolution before going live.

Parameter	Value
Client	Ravi Mareedu & Sudhir Vyakaranam
Instruments	UPRO (3x S&P 500 ETF), TQQQ (3x QQQ ETF)
Data Source	DataBento production daily OHLCV bars
Date Range	Oct 2020 – Dec 2025 (~1,308 trading days)
Initial Capital	\$100,000
Backtest Engine	IFA event-driven backtester
Prepared By	Insight Fusion Analytics (IFA)
Audit Date	April 2026
Sources	feb10_strategy_discussion.srt mar1_meeting.srt

Step 1 — 4-State Machine (UPRO Only)

Step 1 trades UPRO (3x S&P 500) only, using a 4-state machine. Ravi confirmed at mar1 00:23:46: "Bull would be 100% you pro. There's no TQQ... Just do pure three signals, RSI, 200, 50 day average."

State Definitions

State	Allocation	Description
BULL_100	■ 100% UPRO	■ All signals green. Source: mar1 00:23:46 — "Bull would be 100% you pro." ■ Implementation uses 99% cap to prevent negative cash — NOT SPECIFIED BY CLIENT (IFA engineering decision)
BULL_TRIMMED	■ 75% UPRO	■ RSI > 75 triggered partial sell. Source: feb10 00:09:38, mar1 00:27:22
DEFENSIVE	■ 50% UPRO	■ SPY below 50-day SMA. Source: feb10 00:11:24 — "50-Day Moving Average, has breached to the downside. So you're only 50% in." ■ 2-consecutive-day entry confirmation — NOT SPECIFIED BY CLIENT (IFA assumption)
CASH	■ 0%	■ Kill switch active (VIX > 30). Source: feb10 00:10:04-09 — "VIX would be the kill switch." ■ 'OR SPY < 200-SMA' as secondary kill switch — NOT CONFIRMED BY CLIENT. Ravi only mentioned VIX as the kill switch trigger.

Signal Priority (Hierarchy)

■ CONFIRMED — feb10 00:10:51-56: "first hierarchy is Kill Switch, second hierarchy would be SMAs, the third hierarchy would be RSI."

Priority	Signal	Rule
1 (HIGHEST)	KILL SWITCH	■ VIX > 30 → CASH immediately. Source: feb10 00:10:04-09 ■ Whether SPY < 200-SMA is also a kill switch trigger — NOT SPECIFIED BY CLIENT
2	SMA DEFENSIVE	■ SPY < 50-day SMA → trim to 50%. Source: feb10 00:11:24 ■ Entry requires 2 consecutive days below 50-SMA — NOT SPECIFIED BY CLIENT (IFA assumption) ■ Exit from DEFENSIVE also requires 2 consecutive days above — NOT SPECIFIED BY CLIENT
3	RSI OVERBOUGHT	■ RSI > 75 → trim to 75% (BULL_TRIMMED). Source: feb10 00:09:38, 00:31:35 ■ RSI lookback period: 14-day — NOT SPECIFIED BY CLIENT (industry standard) ■ RSI rebuy threshold (RSI < 60 → return to BULL_100) — NOT SPECIFIED BY CLIENT (IFA assumption)
4 (LOWEST)	RE-ENTRY from CASH	■ SPY > 200-SMA → check VIX. Source: mar1 00:10:18 — "once a spy is met more than 200, then you check the results." ■ CONFLICT: VIX re-entry threshold — mar1 00:10:18 says "less than 30"; mar1 00:13:50 says "Wix less than 60". MUST confirm with Ravi before going live.

Step 2 — 6-State Machine (UPRO + TQQQ)

Step 2 adds TQQQ (3x QQQ) as a second sleeve with a 75/25 allocation split. ■ CONFIRMED — mar1 00:09:02: "Remember, we are 75% Spy product and 25% is QQQ product. Spy product is Upro."

SPY signals control the UPRO sleeve. QQQ signals control the TQQQ sleeve independently. Kill switch affects both. ■ CONFIRMED — mar1 00:11:51-12:00: "SMA 50% UPRO, yes, QQQ less than 50, Defensive Trim 50% DQQ, both."

■ State-based logic confirmed — mar1 00:15:24-30: "Apply state-based logic and not trim it next day also."

State Definitions

State	Allocation	Description
BULL_100	■ 75% UPRO + 25% TQQQ	Both SPY and QQQ signals bullish, VIX below threshold.
BULL_TRIMMED	■ 75% of BULL_100 each sleeve	RSI > 75 triggered trim. Both sleeves trimmed proportionally. ■ Whether each sleeve is trimmed independently — NOT SPECIFIED BY CLIENT (IFA assumption)
DEFENSIVE_SPY	■ 50% UPRO sleeve / TQQQ intact	SPY below 50-SMA. UPRO sleeve halved. TQQQ unaffected. ■ Source: mar1 00:11:51
DEFENSIVE_QQQ	■ UPRO intact / 50% TQQQ sleeve	QQQ below 50-SMA. TQQQ sleeve halved. UPRO unaffected. ■ Source: mar1 00:11:51
DEFENSIVE_BOTH	■ 50% each sleeve	Both SPY and QQQ below their 50-SMA. Both sleeves halved.
CASH	■ 0% / 0%	Kill switch active (VIX > 30). Both sleeves liquidated.

Execution Rules

T+1 Execution

■ CONFIRMED — feb10 00:12:31-42: "end of the day is reasonable... next day would be like 30 minutes after opening."

Signal generated at market close (4 PM ET). Trade executed at next day's open, approximately 30 minutes after open. Implementation uses a pending_trade queue — NOT a price column shift. No look-ahead bias.

Commission & Slippage

Parameter	Value / Status
Commission	■ NOT SPECIFIED BY CLIENT IFA used: IBKR tiered (\$0.005/share, min \$1.00, max 1% of trade). Confirm with Ravi.
Slippage — Normal	■ NOT SPECIFIED BY CLIENT IFA used: 5 bps (0.05%). Confirm with Ravi.
Slippage — Stress	■ NOT SPECIFIED BY CLIENT IFA used: 20 bps when VIX $\geq$ 25 or kill switch fires. Confirm with Ravi.
Allocation cap	■ NOT SPECIFIED BY CLIENT IFA used: 99% (not 100%) to prevent negative cash balance. Engineering decision.
Risk-free rate	■ NOT SPECIFIED BY CLIENT IFA used: 4% annual for Sharpe/Sortino. Confirm with Ravi.

Split Adjustments

DataBento data is NOT split-adjusted in CSV files. Splits are applied in-memory during backtest execution. All pre-split OHLC prices are divided by the split ratio.

Instrument	Ratio	Date
UPRO	3:1	2018-05-24
UPRO	2:1	2022-01-13
TQQQ	2:1	2021-01-21
TQQQ	2:1	2022-01-13
TQQQ	2:1	2025-11-20

Strategy 3 — Safety Valve / Predatory Short (SPXU)

■ **IMPORTANT:** Strategy 3 was mentioned in both transcripts but Ravi explicitly deferred it. **No parameters for Strategy 3 were confirmed in any meeting.**

■ **INSTRUMENT CONFIRMED** — mar1 00:35:19: *"In the messages, what is sent you is SPXU"* (SPXU = 3x inverse S&P 500 ETF)

■ **TRIGGER DIRECTION CONFIRMED** — mar1 00:38:10-27: *"VIX is more than 30, you enter XBXU [SPXU]... 50% of the cash you should put in XBXU."*

■ **ALLOCATION CONFIRMED** — mar1 00:38:10-27: 50% of cash to SPXU when VIX > 30.

■ **DEFERRED** — mar1 00:38:32: *"But that's a strategy number three, don't worry, we'll do it later."*

Parameter	Status / Value
SPXU trigger	■ VIX > 30 → enter SPXU with 50% of cash. Source: mar1 00:38:10-27
SPXU instrument	■ SPXU (3x inverse S&P 500). Source: mar1 00:35:19
SPXU allocation	■ 50% of cash position. Source: mar1 00:38:10-27
SPXU exit rule	■ NOT SPECIFIED BY CLIENT — IFA assumption: exit when VIX < 30. Confirm.
SVIX (Short VIX)	■ NOT MENTIONED IN ANY TRANSCRIPT — IFA assumption only. Confirm if required.
SGOV (T-Bills)	■ NOT MENTIONED IN ANY TRANSCRIPT — IFA assumption only. Confirm if required.
Implementation status	DEFERRED — Ravi said 'don't worry, we'll do it later.' Strategy 3 is NOT included in current backtest results.

Claimed Results

Step 1 Results (UPRO Only)

Metric	Value
Final Value	\$274,821
Total Return	174.82%
CAGR	21.44%
Sharpe Ratio	0.676
Sortino Ratio	0.88
Max Drawdown	-38.34% (Mar 10, 2023)
Total Trades	67
Alpha vs SPY	+7.38%

Step 2 Results (UPRO + TQQQ)

Metric	Value
Final Value	\$278,744
Total Return	178.74%
CAGR	21.77%
Sharpe Ratio	0.663
Sortino Ratio	0.861
Max Drawdown	-39.24% (Mar 10, 2023)
Total Trades	180 (90 UPRO + 90 TQQQ)
Alpha vs SPY	+7.71%

Critical Flags — Must Confirm with Ravi Before Going Live

The following parameters were NOT specified by Ravi in either transcript. IFA used industry-standard defaults. These must be confirmed before the strategy is considered production-ready.

#	Type	Parameter	Detail
1	■ CONFL ICT	VIX re-entry threshold	mar1 00:10:18 says VIX < 30 to re-enter. mar1 00:13:50 says VIX < 60 to go to BULL. Which is correct? IFA used < 30. Confirm.
2	■ FLAG	Kill switch — OR condition	Current code uses VIX > 30 OR SPY < 200-SMA as kill switch. Ravi only confirmed VIX > 30 as kill switch. Should SPY < 200-SMA also trigger full CASH? Confirm.
3	■ FLAG	RSI lookback period	NOT SPECIFIED. IFA used 14-day RSI (industry standard). Confirm.
4	■ FLAG	RSI rebuy threshold	NOT SPECIFIED. IFA used RSI < 60 to return from BULL_TRIMMED to BULL_100. Ravi confirmed RSI > 75 triggers trim but said nothing about rebuy. Confirm.
5	■ FLAG	Defensive entry confirmation	NOT SPECIFIED. IFA used 2 consecutive days below 50-SMA to enter DEFENSIVE. Ravi said 'SPY close below 50-day' with no mention of confirmation period. Confirm.
6	■ FLAG	Defensive exit rule	NOT SPECIFIED. IFA used 2 consecutive days above 50-SMA to exit DEFENSIVE. Ravi said nothing about exit from DEFENSIVE. Confirm.
7	■ FLAG	Commission structure	NOT SPECIFIED. IFA used IBKR tiered (\$0.005/share, min \$1, max 1%). Confirm.
8	■ FLAG	Slippage model	NOT SPECIFIED. IFA used 5 bps normal / 20 bps stress. Confirm.
9	■ FLAG	SVIX / SGOV in Strategy 3	NOT MENTIONED IN ANY TRANSCRIPT. IFA added these as assumptions. Ravi only confirmed SPXU. Are SVIX or SGOV required? Confirm.
10	■ FLAG	Strategy 3 exit rule	NOT SPECIFIED. IFA assumed exit SPXU when VIX < 30. Ravi said nothing about exiting the SPXU position. Confirm.

Known Issues & Disclosures

#	Issue	Details
1	Kill switch whipsaw	Mar-Apr 2022 had 4 round trips due to VIX oscillating around 30. No confirmation period on VIX leg — Ravi confirmed VIX is an immediate kill switch (no delay).
2	2022 drawdown	-38% max drawdown during 2022 bear market. Kill switch limited exposure but did not eliminate losses due to T+1 execution gap.
3	T+1 gap risk	Trade executes 30 minutes after open (per Ravi's instruction). Gap between close signal and open execution is not hedged.
4	DataBento split adjustment	In-memory only. Raw CSV files are NOT adjusted. An auditor re-running the code will get the same results; inspecting raw CSVs will show unadjusted prices.
5	Unconfirmed parameters in results	9 of 14 parameters in the backtest were NOT explicitly specified by Ravi. Results are valid for the IFA default assumptions. Changing any flagged parameter will alter results.

— End of Strategy Specification —

Prepared by Insight Fusion Analytics | April 2026 | All parameters sourced from recorded client meetings